GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Digital Circuits — Official GATE ECE Syllabus

Paper Code: EC • Section 5: Digital Circuits (8 - 10 Marks)

Weightage: 8 - 10 Marks • **Subject Code:** DC • **Key Focus:** Combinational & Sequential Circuits, FSM, ADCs/DACs

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: NUMBER SYSTEMS & BOOLEAN ALGEBRA

- Number systems: Binary, octal, decimal, hexadecimal. Inter-base conversion algorithms.
- Signed number representations: Signed magnitude, 1's complement, and 2's complement representations. Range of n-bit signed integers (-2^(n-1) to +2^(n-1) - 1).
- 2's complement arithmetic, addition, subtraction, arithmetic overflow conditions (overflow occurs when sign of operands differs from sign of result).
- Binary codes: BCD (8421 code), Excess-3 code, Gray code (reflected binary code, unit distance property, binary-to-Gray conversion), error detecting and correcting codes (parity bits, Hamming codes).
- Boolean algebra: Fundamental postulates, Huntington's postulates, duality principle, De Morgan's theorems, Shannon's expansion theorem.
- Standard canonical forms: Sum of Products (SOP) and Product of Sums (POS), minterms and maxterms.
- Logic minimization: Karnaugh Map (K-Map) method for 2, 3, 4, and 5 variables. Implicants, prime implicants (PI), essential prime implicants (EPI), don't care conditions (X).
- Logic gates: NOT, AND, OR, NAND, NOR, XOR, XNOR. Universal property of NAND and NOR gates (minimum gate realizations).
- Logic families: CMOS inverter static and dynamic characteristics, noise margin (NML, NMH), propagation delay, power dissipation (static vs. dynamic P = C*V^2*f), fan-in, fan-out.

SECTION: COMBINATIONAL LOGIC CIRCUITS

- Arithmetic circuits: Half adder, Full adder, Half subtractor, Full subtractor. Ripple carry adder (propagation delay tp = n * t_gate), Carry look-ahead adder (CLA, fast addition, generate and propagate terms).
- Binary parallel subtractors, BCD adder, magnitude comparators (1-bit and multi-bit).
- Multiplexers (MUX): 2:1, 4:1, 8:1, 16:1 MUX architecture. Implementation of boolean functions using multiplexers with and without external gates (Shannon's expansion).
- Demultiplexers and Decoders: 2-to-4, 3-to-8 line decoders. Active high and active low outputs, enable pins. Implementing logic functions using decoders with external NAND/OR gates.
- Encoders: Decimal to BCD encoder, Priority encoders (handling multiple simultaneous active inputs, valid bit indicator).
- Code converters: BCD to 7-segment display decoder/driver, binary to Gray code converter.
- Hazard and glitches in combinational circuits: Static-0 hazard, static-1 hazard, dynamic hazard. Elimination of static hazards using consensus terms in K-Map.

SECTION: SEQUENTIAL LOGIC CIRCUITS

- Latches vs. Flip-Flops: Level-triggered vs. edge-triggered storage elements.

• Basic SR latch (NAND and NOR implementations), race conditions, gated SR latch.

• Flip-flops: JK flip-flop, Race-around condition in level-triggered JK flip-flop (occurs when pulse width tp > propagation delay tff). Solutions: Master-Slave JK flip-flop, edge triggering.

• D flip-flop (delay / data flip-flop) and T flip-flop (toggle flip-flop).

• Excitation tables, characteristic tables, and characteristic equations for SR, JK, D, and T flip-flops: Q(next) = S + R'*Q, Q(next) = J*Q' + K'*Q, Q(next) = D, Q(next) = T xor Q.

• Flip-flop conversion techniques.

• Timing parameters: Setup time (t_setup), Hold time (t_hold), Clock-to-Q delay (t_cq). Maximum clock frequency calculation (f_max = 1 / (t_cq + t_comb + t_setup)), clock skew and clock jitter effects, setup and hold time violations.

• Shift registers: SISO, SIPO, PISO, PIPO. Universal shift registers, bidirectional shift registers.

• Ring counters (mod-n counter with n flip-flops, invalid states) and Johnson (twisted ring) counters (mod-2n counter with n flip-flops, decoding logic).

• Counters: Synchronous vs. asynchronous (ripple) counters. Modulus of a counter (Mod-N), count sequence design using state diagrams and excitation tables. Up/down counters.

SECTION: FINITE STATE MACHINES (FSM) & PROGRAMMABLE DEVICES

• Synchronous sequential circuit design methodology: State diagram, state table, state reduction (equivalent states), state assignment.

• Mealy vs. Moore state machines: Mealy machine (output depends on present state AND current input, faster response, fewer states), Moore machine (output depends ONLY on present state, glitch-free output).

• Sequence detectors: Overlapping vs. non-overlapping sequence detection (e.g., detecting '1011' bit patterns).

• Programmable Logic Devices: PROM (fixed AND array, programmable OR array), PAL (programmable AND array, fixed OR array), PLA (programmable AND array, programmable OR array), FPGA and CPLD basic architectures.

SECTION: DATA CONVERTERS & SEMICONDUCTOR MEMORIES

• Digital-to-Analog Converters (DAC): Binary weighted resistor DAC, R-2R ladder DAC (avoids wide range of resistor values, inverted ladder topology).

• DAC specifications: Resolution (V_LSB = V_ref / (2^n - 1)), accuracy, linearity, settling time, monotonic behavior.

• Analog-to-Digital Converters (ADC): Flash (parallel comparator) ADC (fastest, requires 2^n - 1 comparators), Successive Approximation Register (SAR) ADC (requires n clock cycles, binary search algorithm), Dual-slope integrating ADC (slowest, highest accuracy, immune to power supply fluctuations and 50 Hz noise), Counter-ramp ADC.

• ADC specifications: Quantization error (maximum error = +- 1/2 LSB = +- V_LSB / 2), quantization noise power (sigma_q^2 = delta^2 / 12), Signal-to-Quantization Noise Ratio (SQNR = 6.02*N + 1.76 dB).

• Sample and Hold (S/H) circuit: Aperture time, acquisition time, droop rate.

• Semiconductor memories: ROM, SRAM (6T CMOS cross-coupled inverters, fast, volatile), DRAM (1T-1C structure, high density, requires periodic refresh due to capacitor leakage).

SECTION: MICROPROCESSOR (8085) ARCHITECTURE & PROGRAMMING

• 8085 Microprocessor internal architecture: ALU, accumulator (Reg A), general purpose registers (B, C, D, E, H, L), program counter (PC, 16-bit), stack pointer (SP, 16-bit), temporary registers.

• Flag register: Sign (S), Zero (Z), Auxiliary Carry (AC), Parity (P), Carry (CY).

• Address and Data buses: Multiplexed address/data bus (AD0 - AD7) and demultiplexing using ALE (Address Latch Enable) and 74LS373 latch.

• Instruction set classification: Data transfer, arithmetic, logical, branching, machine control instructions.

• Addressing modes: Immediate, direct, register, register indirect, implicit (implied).

• Instruction cycle, machine cycle (Opcode Fetch, Memory Read, Memory Write, I/O Read, I/O Write), T-states. Timing diagrams.

• Interrupt structure: Hardware interrupts (TRAP, RST 7.5, RST 6.5, RST 5.5, INTR). Priority, vectored vs. non-vectored, maskable vs. non-maskable (TRAP is non-maskable, highest priority, level and edge triggered).

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

- Digital Design — M. Morris Mano & Michael D. Ciletti (Pearson)
- Fundamental of Digital Circuits — A. Anand Kumar (PHI)
- Microprocessor Architecture, Programming, and Applications with the 8085 — Ramesh S. Gaonkar

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

- For SAR ADC with n-bit resolution and clock frequency f_clk: Conversion time = n / f_clk (constant regardless of analog input magnitude).
- Maximum clock frequency with clock skew: T_clk >= t_cq + t_comb + t_setup + t_skew. If violated, data corruption occurs.
- Gray code conversion: G3 = B3, G2 = B3 xor B2, G1 = B2 xor B1, G0 = B1 xor B0.